

Becoming a Fractional CTO

What the role is and challenges that come with it, getting into fractional CTO roles, and lessons learned.



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Q: I've heard the term 'Fractional CTO' more and more. What is this role, how do people move into it, and what are career paths as a 'Fractional CTO'?

With the cost of senior engineers and engineering leaders going up, I've noticed more startups retaining CTOs on a fractional contract basis. In a recent issue, [Daria Shulepova](#), co-founder and recruitment partner at Growth Levers, shared her observations about this role growing among early stage startups. From [The Scoop: the 2022 hiring market, as seen by tech recruiters](#):

"I see a growing demand for fractional CTOs. With fractional CTOs, companies and startups tap into great talent who can help build a strong tech infrastructure without having to pay a full-time salary."

This topic on Fractional CTOs is, indeed, intriguing. I cannot think of anyone better to discuss it than an engineering leader who made it their career.

[Sergio Pereira](#) has worked as a startup CTO since 2014, in full-time CTO jobs and in fractional CTO roles. Over the last few years, Sergio has been a fractional CTO for more than 10 companies across a range of industries, tech stacks, and growth stages.

In this issue, Sergio will cover:

1. What is a fractional CTO?

2. Sergio's first fractional CTO job.
3. Taking the plunge: leaving full-time employment to be a fractional CTO
4. Challenges of fractional roles
5. Getting into fractional CTO jobs
6. Are fractional CTO roles growing in popularity?
7. Lessons learned

With this, over to Sergio.

1. What is a fractional CTO?

Ever since I worked in startups and spent my time in startup incubators and accelerators, I saw mentors and consultants who worked with several startups at the same time. Coming from a full-time job myself, it sounded puzzling at first. But now, it all makes sense.

A [Fractional CTO](#) is someone who does exactly that. A fractional CTO is a CTO who works part time for multiple companies in parallel. This happens especially at startups, who see this engagement model as an opportunity to hire an experienced tech leader, even if they can't afford their full-time salary.

Contrary to other short-term consulting arrangements, a Fractional CTO engagement can be a good fit for both the company and the CTO for the medium-long term, and similar Fractional engagements are becoming popular for CMOs, CFOs, and other leadership roles.

2. My first fractional CTO job

In early 2014, I left a "traditional" big consultancy career at Accenture to join as co-founder & CTO at a friend's startup in the AdTech space. That's how I became a Startup CTO in the first place. We got early traction from some big media companies in Europe and joined a startup accelerator to help us structure our growth, as we had never gone through it before.

We joined StartupBootcamp in Amsterdam in 2015. We had mentorship sessions and Founder's "braintrusts" which resulted in my network growing quite fast during those months. That's where my first fractional CTO clients eventually came from.

During the accelerator program, I became a go-to person for other startup Founders that didn't have a technical co-founder. They'd ask for my help with things like interviewing their first engineer hires, integrating with a tech partner, or sorting out some AWS issue or some bug. I did not charge them anything for this help, as I didn't see any of that as "work". I was just helping my buddies figure things out, just like they'd help me in other domains, such as sales or marketing.

My first "official" fractional CTO client, in 2016, was [PakketMail](#), a startup in the accelerator program, whose Founders (Mark and Jorn) were non-technical and needed my help with their technical challenges. I helped them hire 2 engineers and implement an integration with an important tech vendor. I did it for free at first, but they decided to pay me a few months afterward when they got VC funding.

Shortly after, I had a bigger scope engagement with [510.ventures](#), a venture-building company founded by StartupBootcamp mentors. 510.ventures had just locked in a big revenue deal with a Dutch bank and needed to ramp up an engineering team to fulfill it. I did all the sourcing and interviewing for a team of 9 engineers and helped onboard them in Amsterdam. All of the engineers I hired relocated from different countries, which was a great social experience, too.

3. Taking the plunge: Leaving full-time employment to be a fractional CTO

I had worked for these two clients as one-offs while between jobs and alongside a full-time job. It was a nice extra income, but quite exhausting. I found myself working nights and weekends for these clients, so it wouldn't dent my full-time job. I also rejected a couple of other fractional CTO clients because they'd clearly clash with my ability to have a full-time job.

By the end of 2017, I was feeling burned out from my full-time job and gave the fractional CTO opportunity a shot. Having startup founders reach out to me and ask

for my help made me think there was solid market demand, and I might enjoy the flexibility of working 2 or 3 fractional clients in parallel, instead of having 1 full-time job. I also hoped that I could work fully remote as a fractional CTO and experience the "remote" processes of different companies back then.

I was confident I'd get those clients "pretty easily" but I was wrong! Getting clients was much different than what I had imagined. The inbound requests from my network became less common, and fewer leads would convert to paying clients. Also, the jobs I was hired to do took 2-3 months, so I ended up facing the dreaded empty funnel very soon. Lesson learned: Not having a proper sales process can be daunting!

My first plan to get some sales was twofold. I started pitching myself to agencies and applying to speak at tech summits. This went well, but the client leads I got that way were big enterprises, rather than small startups. I found it was actually much easier to get big companies as clients (e.g., automotive, pharma, banks) since they have the budget for external consultants, and they are already hooked up with agencies and recruiters who can place me there. This type of client was very good for me financially, as I proved to myself that working part time was a viable career. But it wasn't as enjoyable for me as working with startup clients since the scope of work was usually much different than the Startup CTO work I enjoy doing.

At Porsche, there was a standout agreement in that regard, where I was the fractional CTO leading the spin-off of a new startup out of their Innovation Lab in Berlin. This spin-off has since then morphed into [Timeless Investments](#), a platform to invest in collectibles. I had other enterprise engagements with well-known companies like BMW, Roche, and PwC (PricewaterhouseCoopers). Those engagements were more like traditional big company consulting, rather than fractional CTO use cases. I didn't enjoy that type of consulting work, where the scope of work was mostly doing hiring/budget plans, architecture diagrams, and presentations to senior management.

These enterprise engagements were very good financially and helped me extend my runway. But in parallel, I managed to finally ramp up a few startup clients, which was much more enjoyable for me. These startup clients came to me through several sales activities, such as speaking at events and writing online, and by word of mouth. I worked with startups across geographies, in different industries, mostly in very early

stages. I enjoyed those, and my biggest challenge in the early days was finding clients with long-term viability. Most of those initial startups were struggling for money, and I worked with them for just 2-3 months at most. I was craving a longer-lasting relationship where I could provide more enduring value.

My first big success as a fractional CTO

In late 2018, I started working with [DeepCode](#), an automated code review startup that Snyk would acquire two years later. My scope of work there leaned more toward product management than hands-on building technology. I worked 25 hours a week, in collaboration with their very technical founder/CTO. My main job was to help DeepCode develop their powerful AI algorithm from a PhD thesis into a tech product.

I'm really proud of the work I did there for about a year. However, since an acquisition was likely in the near future (an acquisition happened in 2020, when Snyk bought the company), I felt like I was missing something. I had no equity or stock options in the company. I realized that working in a Fractional role means I'm usually an outsider. Doing well or doing badly, I'd send the same invoice at the end of the month. At that point, I longed for the ownership and accountability of being a full-timer again.

Fractional CTO -> Fulltime CTO -> fractional CTO

As a fractional CTO, I've been offered to move up to a full-time CTO role with clients that could pay a full-time salary. For example, this happened at [StudentFinance](#). I had been working as their fractional CTO since the company's inception, in early 2019, and I accepted the offer to become their full-time CTO about 9 months later, as they secured VC funding.

I worked there for another 2 years as a co-founder/CTO. It was a great run. I built the product and team from scratch, up to around 30 people, instilled a strong remote culture, and helped the company grow to several markets where we financed STEM education for thousands of students around the world.

Working full time as part of the founding team is hard work. Despite all the good work we did as a team, all during a global pandemic, I wanted to get back to the more flexible lifestyle that fractional CTO engagements provide.

Since 2022, I have taken only fractional CTO work. This time I'm having much more success in mitigating the challenges that caused me mixed feelings in the past.

4. Challenges of fractional roles

The biggest challenge for being happy as a fractional CTO is dealing with the variable nature of revenue. One month I'm juggling 5 clients and earning record revenue. Another month I have no clients and see no new opportunities coming in.

Earnings can be unpredictable and seemingly random at times. This is especially true when my engagements are shorter term. The unpredictable nature of finances caused anxiety for me, and others I know have similar worries. I have done a few things recently to find clients at a more predictable pace:

- **Have cushy savings.** Just like for any other non-deterministic source of revenue, it helps a lot to have more than a one-year runway to account for the rainy days.
- **Create an inbound stream of opportunities.** I hooked myself with "lead generators", ranging from formal (agencies, consulting marketplaces, etc.) to informal (other fractional CTOs that offload excess demand they have). This drives most of my inquiries, and I'm happy to pay them a cut of my earnings (some don't want anything, though).
- **Write in public.** Writing daily on [Twitter](#) and [Medium](#), and frequently on some private groups and roundtables, has gained me a growing number of organic inquiries. It makes me feel more in control and incentivizes me to continue writing consistently in the first place. Plus, I have much better negotiation power toward these clients that reach out to me directly.
- **Identify my target client and build a lead database.** I started using Google Alerts to spot Fintech startups raising seed funding rounds (my target client) each week and store them in a database. I have a few hundred relevant leads to reach out to if I face an empty funnel ahead.

Managing time, given the non-constant and non-deterministic workload, is another common challenge. Sometimes I've been swamped, and other times I've had

an empty funnel. It's also important to allocate a certain bandwidth (say 5-10% of the time) to business development.

Here are a few things I did to manage my time better:

- **Monthly retainers.** Instead of working purely hourly engagements, I prefer selling monthly retainers. This helps with reserving a fixed bandwidth per client, and the constant cash flow it provides helps relieve the pressure to overbook myself into multiple hourly engagements in parallel. Easier said than done, I know!
- **Plan upfront.** I define business development activities clearly and allocate bandwidth to do them, whether it's writing on Twitter, speaking at relevant events, reaching out to prospective clients, or anything else. I prioritize it and learn what works for me. Otherwise, I usually get anxious with a possible empty funnel ahead.
- **Define my target client.** Over the years, I realized that my ideal client is a Fintech startup that has just raised a Seed funding round, operates fully remote, and doesn't have a CTO. Defining this clearly has helped me optimize my business development bandwidth and has also reduced the cases of taking clients I wouldn't enjoy working with.
- **One day, one client.** Several fellow fractional CTOs work with different clients on different days. Others use different laptops or OSs for each client. I never felt such a need to force context switching, but everyone is different.

This way, a client usually reserves a certain number of hours of my time per week or per month. These hours are usually distributed between meetings (which I try to reduce to a bare minimum) and time to work on agreed-on deliverables. As such, my typical week has a small number of meeting slots booked, and some bulky slots reserved for deep work.

No equity. Finally, as I mentioned above, it was a challenge that I didn't have equity in my clients. As such, if a client has clear potential to grow and opportunity for long-term engagement, I try to include some equity in the compensation package.

5. Getting into fractional CTO jobs

Typical fractional CTO engagements tend to exist in a startup context. The scope of work depends a lot on the company's stage, the team, and the specific challenges. Being a fractional CTO is very similar to being a full-time CTO, except that the scope of work is only a subset of what you'd find in a full-time role. Typically you're asked to focus on a company's specific problems, such as hiring the team or building a product:

Pre-seed companies usually need a builder and a coder. If you are a senior software engineer who has previously worked at an early stage startup, you might be a great fit. Even if you've never had a 'CTO' title before, you can come in, roll up your sleeves, and deliver. Every time I have such a hands-on scope of work, I leverage my experience as a software engineer writing code for many years, but I've had fewer such engagements recently.

Between Seed and Series B, it's usually a mix of product work and hiring. My sweet spot is at this stage. As the company secures Seed funding, the usual scope of work is to transition from MVP to scalable product and technology. Another frequent requirement is to move from outsourcing to in-house teams. So I usually need to work with business stakeholders, make technical decisions, and manage the development team.

I have been a CTO in these exact stages for the past 9 years, and my experience helps me stand out with clients that look for CTOs who have experienced a similar growth stage before. They want a CTO that has had experience with early stage startups, going through challenges relevant to the client's scope of work in order to be hired at this stage. Alternatively, if you were an engineering manager, or tech lead in a very relatable company/context, you might be a good fit as well.

Later stage startups are usually about scaling teams and processes. I have observed fewer opportunities in these stages since companies already have the budget for a full time-CTO.

Fellow fractional CTOs who work in such stages have shared how the scope of work is usually around scaling teams and processes. This type of work is not as tech-focused as in the earlier stages. Just like in full-time CTO roles in later stage startups, clients look for fractional CTOs who have experienced blitzscaling or hypergrowth in a

relatable context, and they are willing to pay a premium for that experience. It's also more common for such fractional CTOs to work in parallel to existing full-time CTOs, splitting the work in a way that makes sense for the company.

6. Are fractional CTO roles growing in popularity?

I've observed that demand for part-time roles seems to be growing globally, and not just for fractional CTOs. I hear from friends in Marketing and Finance that fractional CMO and fractional CFO roles are already more "mature" than fractional CTO roles, and there's more liquidity. As such, the number of clients wanting to hire fractional CTOs is also growing fast.

The market for fractional CTOs is mostly composed of startups that want to hire an experienced CTO, but can't afford the salary of a full-time one. So they either reduce the CTO scope of work to focus on a specific problem set, or they make it a hands-off strategic role, which both fit a fractional CTO engagement.

Competition is growing among fractional CTOs, and that's actually very good.

Being a fractional CTO is becoming a thing. Many fellow CTOs are joining private communities that used to be more quiet, such as the [Rands Leadership Slack](#) (find me and other fractional CTOs in the #fractional-cto-vpe channel). Lots of folks are curious about transitioning from full-time to part-time employment across the board (fractional CTO is just one specific role of a much broader trend toward working part time). I receive [Twitter](#) DMs asking about this every week.

A community is building up, where fellow fractional CTOs share experiences and challenges, and offload client leads to one another. I feel the community is maturing, and some standards could eventually emerge, like for contract templates, rate cards, scope of work, and so on. These days, there are roundtables just for fractional CTOs, as a spin-off from traditional CTO roundtables, which is amazing. The community being more active is a lot of fun with this role.

The rapid growth of fractional CTO supply poses a clear pressure on rates. The rapid growth in CTOs, engineering managers, and senior engineers wanting to take on fractional CTO roles often means several people competing for each fractional CTO

opportunity. This pushes everyone to specialize, find their niche client, and invest in their own branding and distribution so that clients find them.

7. Lessons learned

Working as a fractional CTO has clear benefits and some drawbacks.

In my case, the main benefits are the flexible lifestyle and the attractive rates. It's also an arrangement that has accelerated my personal growth as I've been able to work with a broad range of companies, industries, and tech stacks.

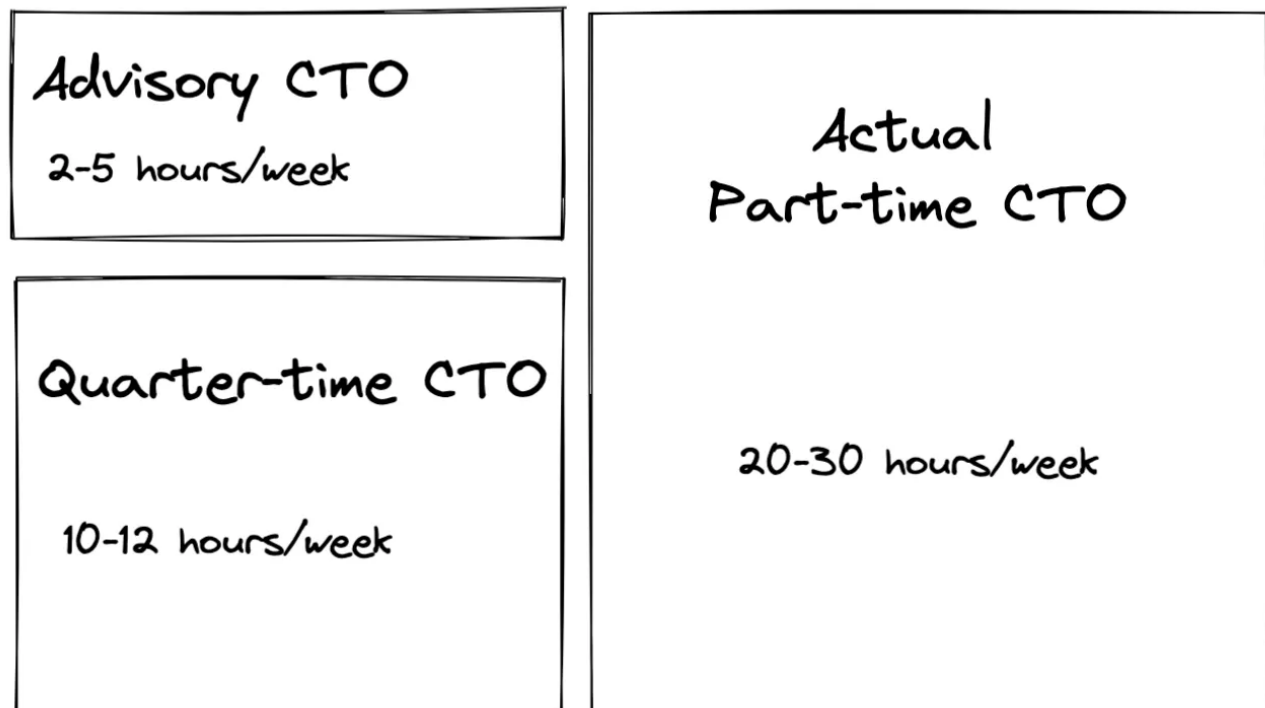
On the flip side, many clients will hire a fractional CTO when things are falling apart. I found myself managing messy situations in several engagements. I also have mixed feelings about the lack of ownership + accountability. At times, I enjoy the detachment, but other times I miss having skin in the game.

How much do fractional CTO roles pay? I've learned an important lesson about rates and negotiating how much to charge each client. There's no standard, but I'd say the most common rates charged to well-funded early stage startups are **between \$150 and \$250 per hour**. To increase rates, you usually need to differentiate via specific knowledge. This knowledge could be a specific tech stack, specific industry requirements, a specific company stage, and so on.

For me, my specific knowledge is having worked with multiple early stage Fintech startups for the last 7 years and being an "early adopter" of remote work. At this moment, I prioritize clients for whom I can leverage this knowledge because I can deliver even greater impact for these clients.

There are 3, distinctively different types of fractional CTO engagements. These are the ones I observed:

- Advisory CTO (2-5 hours per week)
- Quarter-time CTO (10-12 hours per week)
- Actual part-time CTO jobs (20-30 hours per week)



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Types of fractional CTO engagements.

I realized that my preferred arrangement is having one "part-time CTO" client and another 1-2 "advisory CTO" clients on top of it. It all depends on the inquiries I get and the clients I want, but this is how I steer the arrangements to fit.

Takeaways

This is Gergely again.

Thank you, Sergio, for sharing your hard-earned experience on what it's like to be a fractional CTO. Sergio is building [Remote Work Academy](#), where he [writes more](#) about remote work and the fractional CTO role. I also suggest you follow him [on Twitter](#) and [on LinkedIn](#) where he frequently shares insights on these topics.

What does the day-to-day of a fractional CTO look like? This was a question I pushed Sergio on, and he touched on this in the Challenges of fractional CTO jobs, where he detailed how clients reserve a certain number of hours per month.

What I gathered from Sergio is that there is no "typical" day in a Fractional role, similar to how there is no "typical" day in an engineering manager role. There could be meetings, deep work, or solving an unexpected challenge. Managing your own time is a challenge, and it's especially tricky to do when you have more work to do than the hours that the client paid for.

The 3 types of fractional CTO models I found very useful. Advisory (2-5 hours per week), quarter-time (10-12 hours per week), and actual part-time (20-30 hours per week) are all different in commitment and impact.

I'm glad Sergio assigned actual hours to these numbers because it clarifies how it's not realistic to have someone work on two fractional CTO roles in the same sitting (40-60 hours per week, plus context switching). Instead, taking on multiple quarter-time and advisory roles is more realistic, for anyone who is looking to stack these.

The stage of the startup makes a huge difference in the skill sets a company wants. Early stage companies want builders and coders as fractional CTOs who will get the product — or idea — off the ground. For Series A and beyond, hiring and building the organization, or solving specific pain points, become more important.

Later stage companies will rarely, if ever, hire a CTO when the main goal is for them to write code. A CTO should understand the technology and codebase, and be able to help solve persistent problems due to technology (for example, resolving pain points caused by poor tech choices).

I found it interesting how, at later stage companies, some Fractional roles could be hired to work alongside existing CTOs, akin to a consultant.

Fractional CTO roles are not for everyone - but it's hard to tell if this is the case, if you have not yet tried them. [Nicky Wrightson](#) is currently Head of Engineering at [Topi](#), and a former principal engineer at the Financial Times and Skyscanner. After leaving Skyscanner, she took on a fractional CTO role at a venture builder. I asked Nicky about what experience she had with the role. She said she disliked the setup, and this role was not for her. Here's how she put her experience:

"I didn't like how people often wanted me to overcome certain organizational issues, instead of fixing the root cause of these organizational issues.

I was not part of the leadership team but was parachuted into this group. There, I was expected to provide the 'magical technical problem solving ability' that would solve the woes the company has. Even more frustrating, I was sometimes expected to solve for complex problems without collaborating with others.

My experience of being a fractional CTO felt akin to a technical consultancy. It was not the setup I was expecting. I'm glad I did it, but the fractional CTO role is not for me."

The demand for fractional CTOs seems to be going up, but so is the supply. While the idea of part-time technology leadership or part-time tech consulting is not new, it seems there's more talk about fractional CTOs than I heard before. Sergio confirmed this observation as well, as has tech recruiter Daria Shulepova.

It was interesting to hear from Sergio how the supply is changing. To me, a Fractional role sounds like an exciting and interesting one. It's interesting because it offers a relatively risk-free way for experienced software engineers and leaders to get involved with startups, part time, while also getting paid!

I hope you found the overview of this role useful. What other observations have you had while being, or working as, a fractional CTO? Share in the comments below.



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A guest post by

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📢 Writing at <https://twitter.com/SergioRocks>

12 Comments



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William Hurst Jan 10 ❤️ Liked by Gergely Orosz

I am also a fractional CTO operating in Amsterdam. This is a great summary.

It's worth reiterating that you are selling yourself when trying to attract clients. This isn't always comfortable for tech people, but it's a very useful life skill to have.

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Vidal Graupera Writes Engineering Manager Interview Jan 10 ❤️ Liked by Gergely Orosz

For those who might be curious to learn more. The following video, featuring Morgan Craft, a fractional CTO, also discusses this, and explains what a fractional CTO is.

<https://youtu.be/-qssg32FjD0>

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